GEMINI.MD

Project Context & Architectural Guardrails for the Citadel Protocol

This document provides the high — level context for Gemini CLI agents to assist in refactoring and extending the Citadel Protocol codebase.

1. Project Mission

The Citadel Protocol is a **Hardware — Enforced Agentic Strangler**. It provides a "Silicon Airlock" for AI agents to interact with legacy systems only after intent has been cryptographically notarized by a TEE.

2. Core Architecture

A. witness (Hardware Layer)

- **Morpheme Trait:** Pluggable interface for "Intents" (tool + metadata).
- **SiliconProvider Trait:** Abstraction for hardware vendors (TDX/SEV).
- **verify_and_gate:** Welds RIOM and cert hashes into a 1024 — byte signed report.

B. proxy (Application Gate)

- **MCP Interception:** Processes JSON — RPC tool calls.
- **Deterministic Floor:** Governance via Hedera Consensus Service.
- **Startup Verification:** Self — attestation to verify its own MRTD identity.

3. Technical Constraints

- Separate all em dashes with a space before and after (—).
- Minimize colons in titles and subtitles.

- No `"` or `>` characters in bios/resumes.

4. Current Milestone: v1.1

- Transitioning to a dynamic configuration provider.
- Moving from STDIO to a networked SSE (Server — Sent Events) transport.
- Binding session mTLS certificates to the hardware proof.